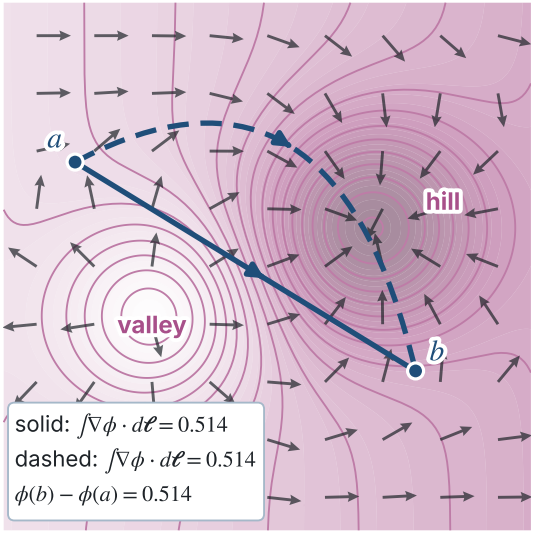
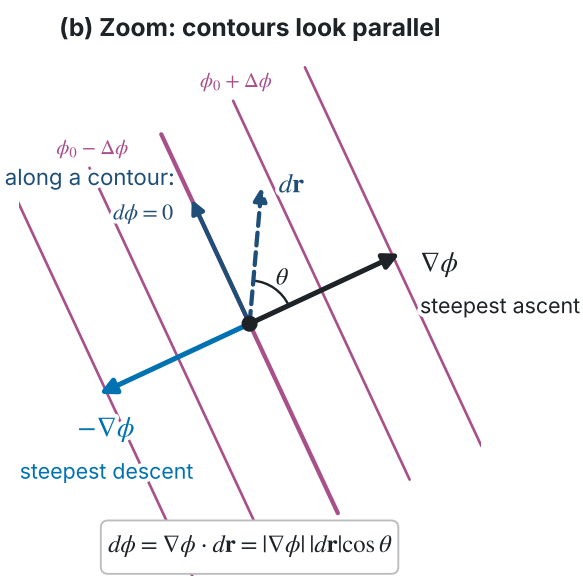

(a) Contours and gradient (uphill arrows)
a
hill
valley
b
solid: $\int \nabla\phi \cdot d\boldsymbol{\ell} = 0.514$
dashed: $\int \nabla\phi \cdot d\boldsymbol{\ell} = 0.514$
$\phi(b) - \phi(a) = 0.514$


(b) Zoom: contours look parallel
$\phi_0 + \Delta\phi$
$\phi_0 - \Delta\phi$
along a contour:
$d\phi = 0$
$d\mathbf{r}$
θ
$\nabla\phi$
steepest ascent
$-\nabla\phi$
steepest descent
$d\phi = \nabla\phi \cdot d\mathbf{r} = |\nabla\phi|\,|d\mathbf{r}|\cos\theta$